

Python Programming – Solutions

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(Without shortcut built-in functions such as max(), isupper(), islower())

Q1. Reverse a 2-digit number

Code

```
# Reverse a 2-digit number

num = int(input("Enter a 2-digit number: "))

ones = num % 10
tens = num // 10

reversed_num = ones * 10 + tens

print("Reversed number:", reversed_num)
```

Sample Output

```
Enter a 2-digit number: 47
Reversed number: 74
```

Q2. Check the case of a character

Code

```
# Check whether a character is uppercase, lowercase, or not a
letter

ch = input("Enter a character: ")

if 'A' <= ch <= 'Z':
    print("It is an UPPERCASE letter")
elif 'a' <= ch <= 'z':
    print("It is a lowercase letter")
else:
    print("Not an alphabet character")
```

Sample Output

```
Enter a character: g
It is a lowercase letter
```

Q3. Maximum among three numbers

Code

```
# Find the maximum among three numbers

a = float(input("Enter first number: "))
b = float(input("Enter second number: "))
c = float(input("Enter third number: "))

largest = a

if b > largest:
    largest = b

if c > largest:
    largest = c

print("Maximum number is:", largest)
```

Sample Output

```
Enter first number: 5
Enter second number: 9
Enter third number: 3
Maximum number is: 9.0
```

Q4. Factorial of a number

Code

```
# Factorial of a number

n = int(input("Enter a number: "))

fact = 1

for i in range(1, n + 1):
    fact = fact * i

print("Factorial of", n, "is", fact)
```

Sample Output

```
Enter a number: 5
Factorial of 5 is 120
```

Q5. GCD of two numbers

Code

```
# GCD using Euclidean algorithm

a = int(input("Enter first number: "))
b = int(input("Enter second number: "))

while b != 0:
    remainder = a % b
    a = b
    b = remainder

print("GCD is:", a)
```

Sample Output

```
Enter first number: 36
Enter second number: 60
GCD is: 12
```

Q6. Prime or Composite

Code

```
# Check whether a number is prime or composite

n = int(input("Enter a number: "))

if n < 2:
    print(n, "is neither prime nor composite")
else:
    count = 0

    for i in range(1, n + 1):
        if n % i == 0:
            count = count + 1

    if count == 2:
        print(n, "is Prime")
    else:
        print(n, "is Composite")
```

Sample Output

```
Enter a number: 17
17 is Prime
```

Q7. Pattern — A A A A / B B B / C C / D

Code

```
# Letter Pattern

n = 4

for i in range(n):
    letter = chr(65 + i)
    row = (letter + " ") * (n - i)
    print(" " * i + row.rstrip())
```

Sample Output

```
A A A A
 B B B
  C C
   D
```

Q8. Centered Number Triangle

Code

```
# Centered Number Triangle

n = 3

for i in range(n):
    num = i + 1
    spaces = (n - i - 1) * 2

    print(" " * spaces, end="")

    for j in range(2 * i + 1):
        print(num, end=" ")

    print()
```

Sample Output

```
  1
 2 2 2
3 3 3 3 3
```
